Time Value of Money

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Learning Outcomes

- Understand that money today is more valuable than the same amount in the future.
- Differentiate between present value (PV) and future value (FV).
- Apply TVM concepts to real-life financial decisions like saving, investing, or borrowing.
- Calculate PV and FV of single sums, annuities, and perpetuities.
- Understand and compute compound interest and the effect of compounding frequency.
- Use discounting to evaluate future cash flows and investment opportunities.
- Build a foundation for advanced topics in finance, such as valuation and risk analysis.

Why is money today more special than money tomorrow? Is it because money likes to time travel?

Time Value of Money (TVM)

Time Value of Money (TVM)

- The Time Value of Money (TVM) is one of the most fundamental concepts in finance and economics.
- It reflects the idea that a rupee today is worth more than a rupee in the future due to its potential earning capacity.
- This core principle forms the basis for evaluating investments, loans, annuities, and virtually all financial decisions.

Key Components of Time Value of Money

1. Present Value (PV): Value of future money in today's terms.
2. Future Value (FV): Amount of money money will grow to in the future with interest.
3. Interest Rate ®: Rate at which money earns or is discounted.
4. Time Period (n): Number of periods (years/months) involved.
5. Payments (PMT): Regular cash inflows or outflows (e.g., loan installments).
6. Compounding Frequency: How often interest is added (e.g., annually, monthly)

Features of Time Value of Money:

- Money has time-based value.
- Future money is worth less than present money.
- Interest can be simple or compound.
- Time Value of Money (TVM) applies to single sums or series of cash flows.
- Compounding increases value over time.
- Discounting reduces future value to present.

Importance of Time Value of Money

- Helps make better investment decisions.
- Essential for comparing financial options over time.
- Used in loan calculations and repayment planning.
- Crucial for retirement and savings planning.
- Applies in valuing stocks, bonds, and other assets.
- Forms the basis of financial tools like NPV and IRR.

Important Terms in TVM

1. Net Present Value (NPV): Value of future cash flows minus initial investment.
2. Internal Rate of Return (IRR): Return rate that makes NPV = 0.
3. Annuity: Series of equal payments over time.
4. Perpetuity: An annuity that goes on forever.
5. Discount Rate: Rate used to bring future value to present.
6. Compound Interest: Earning interest on both principal and previous interest.

Present Value (PV)

- Present Value (PV) is the current worth of a sum of money or stream of cash flows that you expect to receive in the future, discounted at a specific interest (or discount) rate.

The basic formula:

$$PV = \frac{FV}{(1+r)^n}$$

- PV = Present Value
- FV = Future Value
- r = Interest rate (decimal)
- n = Number of periods

Present Value (PV)

Example: Present Value of a Single Future Amount

- You will receive 1,000 ₹ in 3 years. The discount rate is 5%. What is the present value?

$$PV = \frac{1000}{(1+0.05)^3} = \frac{1000}{1.157625} = 863.84$$

So, 1,000 received in 3 years is worth about 863.84 ₹ today at 5% discount rate.

Future Value (FV)

- The amount a present sum of money will grow over time with compound interest.

The basic formula:

$$FV = \mathrm{PV} \times (1 + r)^n$$

- PV = Present Value
- FV = Future Value
- r = Interest rate (decimal)
- n = Number of periods

Future Value (PV)

- Example: You invest ₹1,000 today at an annual interest rate of 5%. How much will your investment be worth after 2 years if the interest is compounded annually?

$$FV = \mathrm{PV} \times (1 + r)^n$$

Where:

$$\mathrm{FV}=1000 \times (1 + 0.05)^2 = 1000 \times 1.1025 = 1102.50$$

- PV=1000
- r=0.05 (5%)
- n =2 years

After 2 years, your investment will grow to ₹1,102.50.

Compounding

- Compounding is the process in which the interest you earn on a principal amount starts earning its own interest.
- In simple terms: Interest on interest.

How It Works:

- Every period (e.g., year, month), you earn interest on:
- The original principal, and
- The interest previously earned.
- This leads to exponential growth over time.

Types of Compounding

Compounding Frequency	Periods per Year (n)	Example
Annual	1	Once per year
Semi-Annual	2	Twice per year
Quarterly	4	Every 3 months
Monthly	12	Every month
Weekly	52	Every week
Daily	365	Every day (common in banks)
Continuous	∞	Theoretical max compounding

Continuous Compounding

- When compounding happens infinitely many times, it uses the formula:

Basic Formula : $\mathrm{FV} = \mathrm{PV} \times e^{rt}$

- e = Euler's number (≈ 2.71828)
- r = Annual interest rate
- t = Time in years

Applications of Compounding

- Investment growth
- Loan amortization
- Retirement planning
- Business forecasting
- Savings account interest
- Bond pricing

Compound Interest vs. Simple Interest

Feature	Simple Interest	Compound Interest
Interest Earned On	Principal only	Principal + accumulated interest
Growth Pattern	Linear	Exponential
Total Return	Lower	Higher
Example Use Case	Short-term loans	Long-term investments/savings

Annuity

- An annuity is a series of equal payments made at regular intervals over time.
- The payments can be made weekly, monthly, annually, etc.
- Examples: Monthly rent, car loans, mortgage payments, retirement income

Why Are Annuities Important?

- Used in loans, mortgages, leases, pensions, and insurance.
- Help calculate the value of investments with regular cash flows.
- Basis for many financial products and decisions.

Present Value of an Ordinary Annuity (PVOA)

- The Present Value of an Ordinary Annuity is the current (today's) value of a series of equal payments made at the end of each period, discounted at a specific interest rate.

- Basic formula for PVOA : $PV = P \times \frac{1-(1+r)^{-n}}{r}$

Example: What is the present value of an ordinary annuity if you receive ₹1,000 annually for 5 years and the discount rate is 6%?"

$$PV = 1000 \times \frac{1-(1+0.06)^{-5}}{0.06} = 1000 * 4.21236 = 4212.36$$

So, the present value of receiving ₹1,000 annually for 5 years at 6% interest is ₹4,212.36.

Future Value of an Ordinary Annuity (FVOA)

The future value of an ordinary annuity is the total value of all the payments at a future date, after earning interest over time.

- Basic formula for FVOA : $FV = P \times \frac{(1+r)^n - 1}{r}$

Example: What will be the future value of investing ₹1,000 per year for 5 years at an annual interest rate of 6%, compounded annually?.

$$FV = 1000 \times \frac{(1+0.06)^5 - 1}{0.06} = 1000 \times 5.6371 = 5{,}637.10$$

Investing ₹1,000 yearly for 5 years at 6% interest grows to a future value of ₹5,637.10.

Perpetuity

- A perpetuity is a series of equal cash flows (payments or receipts) that:

Key features :

1. Infinite Payments: Perpetuity provides cash flows that continue forever.
2. Equal Intervals: Payments occur at regular, consistent time periods.
3. Fixed Amount: Each payment is the same (in a standard perpetuity).
4. Finite Present Value: Despite infinite payments, the present value is calculable.

Perpetuity

1. Constant (Level) Perpetuity:

- Pays a fixed amount at regular intervals forever.
- No change in payment amount over time.

Formula: $PV = \frac{P}{r}$

Example: What is the present value of a preferred stock that pays a fixed dividend of ₹100 per year forever if the required return is 5%?

$$PV = \frac{100}{0.05} = 2{,}000$$

- Therefore, the present value of this perpetuity (the amount you should be willing to pay today for the preferred stock) is ₹2,000.
- This means if you pay ₹2,000 now, you will earn a 5% return by receiving ₹100 every year indefinitely.

Perpetuity

2. Growing Perpetuity

- Payments increase at a constant rate (g) each period forever.
- Common in stock valuation (e.g., growing dividends).

Formula: $PV = \frac{P}{r-g}$ (only valid when r>g)

Example: What is the present value of a perpetuity that pays ₹50 next year and grows at 3% annually, given a discount rate of 8%?

$$PV = \frac{50}{0.08 - 0.03} = 1000$$

The present value of the growing perpetuity is ₹1000. This value applies because the discount rate (8%) is greater than the growth rate (3%).